🌐 VLANs Part 2 (VLAN Trunking & Inter-VLAN Routing) (10.31.2026)

> **Notion Page Summary:** This note covers the configuration, verification, and purpose of **Trunk Ports** and **Router on a Stick (ROAS)** for inter-VLAN routing. It includes a deep dive into **802.1Q encapsulation**, the **Native VLAN**, and a comprehensive command cheat sheet for Cisco IOS.

📖 1. Core Concepts

🔗 What is a Trunk Port?

An access port belongs to a single VLAN, but a **trunk port** is a point-to-point link that carries traffic for **multiple VLANs** over a single physical interface. It is typically used to connect switches to other switches or to a router (for ROAS).

🎯 Purpose of Trunk Ports

- **Efficiency:** Reduces the need for multiple physical links (one per VLAN).
- **Scalability:** Easily extend VLANs across a network without additional cabling.
- **Inter-Switch Communication:** Allows switches to share VLAN information and forward frames correctly.

🏷️ 802.1Q Encapsulation (The "Tag")

802.1Q tag format

16 bits	3 bits	1 bit	12 bits
TPID	TCI		
	PCP	DEI	VID

To identify which VLAN a frame belongs to, switches use the **IEEE 802.1Q** standard.

- **How it works:** A **4-byte tag** is inserted into the Ethernet frame.
- **Key Field:** The **VLAN ID (VID)** inside the tag (12 bits) specifies the exact VLAN (e.g., VLAN 10, 20, 30).
- **Result:** The receiving switch reads the VID and forwards the frame only to ports in that VLAN.

🏠 The Native VLAN (The "Untagged" Traffic)

This is a crucial 802.1Q feature.

- **What it is:** Traffic from the **Native VLAN** is sent across the trunk **untagged** (no 802.1Q tag is inserted).
- **Why it matters:** It ensures compatibility with older devices that don't understand VLAN tags.
- ⚠️ **Best Practice:** The Native VLAN **must match on both ends** of the trunk. If it doesn't, traffic will be misdirected, causing a **VLAN mismatch** and potential security vulnerabilities.

⌨️ 2. Trunking Command Cheat Sheet (Cisco IOS)

Here are the essential commands for configuring a trunk on a Cisco switch.

🛠️ Configuration Commands

Command (in Interface Config Mode)	Purpose
`switchport mode trunk`	**Forces** the port into permanent trunk mode.
`switchport trunk encapsulation dot1q`	Sets the trunking encapsulation to **802.1Q** (required on some switch models before entering trunk mode) .
`switchport trunk native vlan [vlan-id]`	**Sets the Native VLAN** for the trunk. **Example:** `switchport trunk native vlan 10` makes VLAN 10 the untagged VLAN.
`switchport trunk allowed vlan [vlan-list]`	**Sets the exact list** of VLANs allowed on the trunk (overwrites any previous list). **Example:** `switchport trunk allowed vlan 10,20,30`
`switchport trunk allowed vlan add [vlan-list]`	**Adds** VLANs to the existing allowed list without overwriting it .
`switchport trunk allowed vlan remove [vlan-list]`	**Removes** specific VLANs from the allowed list.
`no switchport trunk allowed vlan`	**Resets** the allowed list to the default, which is **all VLANs** (1-4094) .
`switchport nonegotiate`	Disables DTP (Dynamic Trunking Protocol) negotiation, a security best practice.

🔍 Verification Commands

Always verify your configuration!

Command (in Privileged EXEC Mode)	What It Shows
`show interfaces trunk`	**The most important command.** Displays all active trunks, their mode, encapsulation, **Native VLAN**, and the lists of **allowed and active VLANs** .
`show interfaces [interface-id] switchport`	Provides detailed administrative and operational status of a specific port, including its trunking mode and encapsulation .
`show vlan brief`	Lists all VLANs and their associated access ports.
`show ip interface brief`	(For ROAS) Verifies the IP addresses and status of router sub-interfaces.

🧠 Pro-Tip: Understanding `show interfaces trunk`

The output has three critical VLAN lists:

1. **Vlans allowed on trunk:** The VLANs *configured* to pass. If a VLAN is missing here, it's a configuration error (e.g., you forgot to use the `add` keyword).
2. **Vlans allowed and active in management domain:** The VLANs that are *allowed* AND exist in the switch's VLAN database. If a VLAN is allowed but missing here, the VLAN hasn't been created on the switch .
3. **Vlans in spanning tree forwarding state:** The VLANs that are actually forwarding traffic after STP calculations.

🏢 3. Router on a Stick (ROAS)

ROAS is a technique to enable **inter-VLAN routing** using a single physical router interface.

- **How it works:** The physical router interface is divided into multiple **logical sub-interfaces**, one for each VLAN.
- **The Process:**
  1. The switch port connected to the router is configured as a **trunk**.
  2. Each router **sub-interface** is assigned an **IP address** (acting as the default gateway for that VLAN) and is configured with **802.1Q encapsulation** for its specific VLAN ID .

⚙️ ROAS Configuration on the Router

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0.10   ! Create sub-interface for VLAN 10
Router(config-subif)# encapsulation dot1q 10        ! Assign 802.1Q encapsulation for VLAN 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0  ! Set the gateway IP
Router(config-subif)# exit

Router(config)# interface gigabitethernet 0/0.20   ! Create sub-interface for VLAN 20
Router(config-subif)# encapsulation dot1q 20        ! Assign 802.1Q encapsulation for VLAN 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0  ! Set the gateway IP
Router(config-subif)# exit

Router(config)# interface gigabitethernet 0/0       ! Don't forget to enable the physical interface!
Router(config-if)# no shutdown
```

> **Note:** The sub-interface number does **not** have to match the VLAN ID, but it is a best practice to do so for clarity .

🛡️ 4. Security Best Practices

- **Change the Native VLAN:** Do not use VLAN 1 as the Native VLAN. Use an unused, dedicated VLAN to mitigate **VLAN hopping** attacks .
- **Prune Allowed VLANs:** Instead of allowing all VLANs, explicitly list only the VLANs that need to traverse the trunk using the `switchport trunk allowed vlan` command.
- **Disable DTP:** On trunks, disable Dynamic Trunking Protocol with `switchport nonegotiate` to prevent an attacker from negotiating a trunk.
- **Tag the Native VLAN:** For maximum security, use the global command `vlan dot1q tag native`. This forces the switch to tag even the Native VLAN traffic and drop any untagged frames, preventing VLAN hopping .

✍️ 5. Quiz Answers & Explanations

Here are the answers to your quiz, with a bit more context.

Q1: Configure SW1 to send VLAN10 frames untagged over its GigabitEthernet0/1 interface.

- **Answer: d)** `switchport trunk native vlan 10`
- **Why:** Frames from the **Native VLAN** are sent untagged across an 802.1Q trunk by default .

Q2: After modifying the allowed VLAN list, return it to the default state.

- **Answer: b)** `switchport trunk allowed vlan all`
- **Why:** The `no` form of the command resets the list, but `all` explicitly sets it to allow all VLANs (1-4094) .

Q3: The command `switchport mode trunk` is rejected. What might fix it?

- **Answer: c)** `switchport trunk encapsulation dot1q`
- **Why:** Some Cisco switches require you to specify the encapsulation type **before** the port can be placed into trunk mode .

Q4: Which field of the 802.1Q tag identifies the VLAN ID?

- **Answer: b) VID**
- **Why:** The **VLAN Identifier (VID)** is the 12-bit field that carries the VLAN number.

Q5: You added VLAN10 with `switchport trunk allowed vlan add 10`, but it's missing from "Vlans allowed and active." Why?

- **Answer: a) VLAN10 doesn't exist on the switch.**
- **Why:** For a VLAN to be "active," it must be **created** in the switch's VLAN database. The `allowed` command only permits it; the VLAN must also exist to become active .